Wenkai Qin

CONTACT INFORMATION

Department of Aerospace Engineering and Engineering Mechanics
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EDUCATION

University of Texas at Austin, Austin, Texas USA
Ph.D., Aerospace Engineering, May 2026 (expected)
Advisor: Todd E. Humphreys
Affiliation: Radionavigation Laboratory

University of Texas at Austin, Austin, Texas USA
M.S., Aerospace Engineering, May 2024
Advisor: Todd E. Humphreys
Affiliation: Radionavigation Laboratory
Report: An Agile, Portable Antenna System for LEO Megaconstellation-Based PNT and Timing Properties of the Starlink Ku-Band Downlink: A Preliminary Study

Harvey Mudd College, Claremont, California USA
B.S., Engineering, May 2018
Humanities Concentration: Japanese
Graduated with distinction

ACADEMIC POSITIONS

Research Assistant, University of Texas at Austin — September 2022 – present

OTHER PROFESSIONAL EXPERIENCE

Amazon — May 2025 – August 2025
Applied Scientist — Austin, TX

Designed and prototyped alternative positioning, navigation and timing (PNT) solutions for the Kuiper satellite communications constellation.
Architected and implemented a comprehensive hardware-in-the-loop signal chain to generate Kuiper down-link waveforms, process them through custom space-ground channel emulation, and produce a real-time simulated PNT fix.
Developed initial network-entry algorithms to minimize cold-start acquisition and authentication time for user terminals.
Successfully deployed the complete system stack onto a phased-array-equipped mobile test platform, enabling field validation in dynamic environments.

Raytheon Missiles and Defense — September 2019 – April 2022
Guidance Navigation and Control Engineer II — Tucson, AZ
Guidance Navigation and Control Engineer I

Developed optimal guidance, navigation, and control laws for use on defense interceptors.
Captured and interpreted physical flight test data to improve 6DOF simulation fidelity.
Designed exo-atmospheric pointing algorithm, cutting per-event cold gas usage by over 30%.

Developed autonomous horizon-based navigation algorithm (US Patent No. US11580690B1).

Laserfiche February 2019 – September 2019
Data Engineer Long Beach, CA

Led project to launch and deploy in-house business intelligence data warehouse.
Designed, built, deployed, and maintained enterprise-wide data warehouse, data pipeline, and analytics platform.
Developed data reports, visualizations, and interactive business intelligence reports for key company stakeholders.
Cleaned data, improved data quality through data inaccuracy profiling and recommended process improvements.

C2 Education July 2018 – February 2019
Private Tutor Greenwich, CT

Taught, mentored, and motivated middle and high schoolers in a 3:1 student to teacher environment.
Tracked, analyzed academic progress and tailoring tutoring sessions to each student's personal goals.
Assisted in detailed record-keeping, communicating with parents, and other administrative tasks.
Covered topics in math, science and English, including AP Physics and Calculus, ACT Science, SAT Reading.

MEMBERSHIPS IN PROFESSIONAL AND HONORARY SOCIETIES

Member, Institute of Electrical and Electronics Engineers (IEEE)
Member, Institute of Navigation (ION)

HONORS AND AWARDS

Department of Defense NDSEG Fellowship, 2024 – present

PUBLICATIONS

Refereed Journal Publications

J1 **W. Qin**, A. M. Graff, Z. L. Clements, Z. M. Komodromos, and T. E. Humphreys, "Timing properties of the Starlink Ku-band downlink," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 62, pp. 727–744, 2026 [pdf]

J2 **W. Qin**, M. L. Psiaki, J. R. Bowman, and T. E. Humphreys, "Pilots and other predictable elements in the Starlink Ku-band downlink," *npj Wireless Technology*, vol. 2, no. 1, p. 69, 2026 [pdf]

J3 **W. Qin** and T. E. Humphreys, "T-code reconstruction for low-SNR acquisition of Starlink Ku-band frames," 2026. In preparation.

Refereed Conference Proceedings

C1 Z. M. Komodromos, **W. Qin**, and T. E. Humphreys, "Signal simulator for Starlink Ku-Band downlink," in *Proceedings of the ION GNSS+ Meeting*, pp. 2798–2812, 2023 [pdf]

C2 **W. Qin**, Z. M. Komodromos, and T. E. Humphreys, "An agile, portable antenna system for LEO megaconstellation-based PNT," in *Proceedings of the 36th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2023)*, (Denver, CO), pp. 2835–2846, Sep. 2023 [pdf]

C3 Z. M. Komodromos, **W. Qin**, and T. E. Humphreys, "Weak signal acquisition and tracking of the Starlink Ku-Band downlink to enable global PNT," in *Proceedings of the ION Joint Navigation Conference (JNC)*, 2024

C4 **W. Qin**, Z. M. Komodromos, and T. E. Humphreys, "An analysis of the short-term time stability of the Starlink Ku-band downlink frame clock," in *Proceedings of the IEEE International Conference on Wireless for Space and Extreme Environments (WISEE 2024)*, 2024 [pdf]

C5 S. C. Morgan, Z. M. Komodromos, **W. Qin**, Z. L. Clements, A. M. Graff, W. J. Morrison, and T. E. Humphreys, "A mock implementation of fused LEO GNSS," in *Proceedings of the IEEE/ION PLANS Meeting*, (Salt Lake City, UT), 2025 [pdf]

C6 Z. M. Komodromos, S. C. Morgan, Z. L. Clements, **W. Qin**, W. J. Morrison, and T. E. Humphreys, "Network-aided pseudorange-based LEO PNT from OneWeb," in *Proceedings of the IEEE/ION PLANS Meeting*, (Salt Lake City, UT), 2025 [pdf]

C7 **W. Qin**, Z. M. Komodromos, S. C. Morgan, and T. E. Humphreys, "Maximum likelihood time of arrival and Doppler estimation for precise Starlink-based PNT," in *Proceedings of the IEEE/ION PLANS Meeting*, (Salt Lake City, UT), 2025 [pdf]

Patents

P1 J. N. Head, **W. Qin**, J. S. Klein, and J. D. Verret, "Horizon-based navigation," US Patent 11,580,690 B1, February 2023. [link]

TEACHING ACTIVITIES

1. Spring 2015: Graded and tutored for CS5 (Introduction to Computer Science) at Harvey Mudd College
2. Fall 2015: Graded and tutored for Physics 24 (Mechanics and Wave Motion) at Harvey Mudd College
3. Fall, Spring 2018: Graded and tutored for E101-102 (Advanced Systems Engineering I-II) at Harvey Mudd College